

Generation Interconnection System Impact Study Report

For Queue Project Q23-2309

Old Bridge Clean Energy Center 138 kV

December 01, 2024

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q23-2309. The Interconnection Customer proposes a 259.9 MW solar photovoltaic facility interconnecting at the Old Bridge Clean Energy Center 138 kV substation in the New Jersey area. The request was received on April 26, 2023. The study evaluated system intact and single contingency conditions using the regional planning case, including 517 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$4,449,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Parlin Power Plant Substation-Old Bridge Clean Energy Ctr 138.0 kV line	Old Bridge Clean Energy Ctr-Rutherford 138.0 kV	147.6	221	99.9
Rutherford-Old Bridge Clean Energy Ctr 138.0 kV line	Parlin Power Plant-Old Bridge Clean Energy Ctr 138.0 kV	137.1	238	99.9

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Parlin Power Plant Substation-Old Bridge Clean Energy Ctr 138.0 kV line	Rebuild line with higher-capacity conductor	\$2,127,000	100% (\$2,127,000)
Old Bridge Clean Energy Ctr-Rutherford 138.0 kV line	Rebuild line with higher-capacity conductor	\$2,322,000	100% (\$2,322,000)
	Total allocated to project		\$4,449,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.